

BOWEI LI

(+1) 412-954-7690 boweili@andrew.cmu.edu
Homepage Google Scholar GitHub LinkedIn

EDUCATION

Carnegie Mellon University

Aug 2024 – May 2026

M.S. in Electrical and Computer Engineering (Advanced Study)

GPA: 3.86/4.00

Xidian University & Heriot-Watt University

Sep 2020 – Jun 2024

Joint B.Eng. in Telecommunication Engineering

GPA: 3.80/4.00

RESEARCH INTERESTS

Robot learning for manipulation, with a focus on long-horizon skill composition, agentic systems, and simulation-based data generation for real-world transfer.

PUBLICATIONS & PREPRINTS

*Equal contribution.

Journal, Conference & Workshop Papers

- [1] Z. He, **B. Li**, J. Dou, Y. Zhang, and C. Liu, “Motion Planning with Model-Based Diffusion via Constraint Optimization and Adaptive Scheduling,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. [Paper]
- [2] **B. Li***, P. Yu*, Z. Tang, H. Zhou, Y. Sun, R. Liu, and C. Liu, “NeSyPack: A Neuro-Symbolic Framework for Bimanual Logistics Packing,” *Robotics: Science and Systems (RSS) Workshop*, 2025. [Paper]
- [3] Y. Sun, R. Chen, K. S. Yun, Y. Fang, S. Jung, F. Li, **B. Li**, W. Zhao, and C. Liu, “SPARK: Safe Protective and Assistive Robot Kit,” *IFAC Symposium on Robotics*, 2025. [Paper]
- [4] **B. Li**, S. Tripathi, A. K. M. S. Hossain, R. Zhang, M. Wang, and J. Xie, “When Learning Meets Dynamics: Distributed User Connectivity Maximization in AAV-Based Communication Networks,” *IEEE Transactions on Cognitive Communications and Networking*, 12(1):175–188, 2026. [Paper]
- [5] R. Zhang, **B. Li**, L. Zhang, J. Xie, and M. Wang, “Learning with Dynamics: Autonomous Regulation of UAV-Based Communication Networks with Dynamic UAV Crew,” *IEEE Communications Magazine*, 64(1):72–78, 2026. [Paper]
- [6] **B. Li**, Y. Xu, R. Zhang, J. Xie, and M. Wang, “Maximizing User Connectivity in AI-Enabled Multi-UAV Networks: A Distributed Strategy Generalized to Arbitrary User Distributions,” *IEEE International Conference on Communications (ICC)*, pp. 530–535, 2025. [Paper]

Manuscripts Under Review

- [7] **B. Li**, Y. Zhang, and C. Liu, “ARSTAG: An Agentic Real2Sim2Real System for Task-Specific Robot Data Generation,” *Manuscript*, 2026. Submitted to ICRA.
- [8] J. Yu*, **B. Li***, Z. Tang, G. Lu, C. Hu, R. Liu, and C. Liu, “BrickCraft: Visuomotor Skill Composition with Situated Manual Guidance for Long-Horizon Interlocking Brick Assembly,” *arXiv:2605.07605*, 2026. Submitted to *IEEE Robotics and Automation Letters (RA-L)*. [Paper] [Project]
- [9] H. Hu*, **B. Li***, Z. Wang*, T. Wei, C. Hutchison, E. Sample, and C. Liu, “Enhancing Certifiable Semantic Robustness via Robust Pruning of Deep Neural Networks,” *arXiv:2510.00083*, 2025. Submitted to the *American Control Conference (ACC)*. [Paper]
- [10] P. Yu, M. Dabhi, S. Li, **B. Li**, L. Jeni, and C. Liu, “ManiSkillFormer: Demonstration-Free Compositional Manipulation via Task-Conditioned Geometric Contracts,” *arXiv:2609.16331*, 2026. Submitted to ICRA. [Paper]

RESEARCH EXPERIENCE

Research Assistant, Robotics Institute

Sep 2024 – Present

Advisor: *Prof. Changliu Liu*

Carnegie Mellon University

- **ARSTAG [7] — Submitted to ICRA.** Led method design, implementation, and real-robot evaluation of an agentic Real2Sim2Real system that turns an RGB image and a task instruction into task-specific simulation scenes, demonstrations, and robot policies. Generated **1,000 simulated demonstrations per task across seven tasks**; fine-tuned $\pi_{0.5}$ achieved **74.6% average real-world success** on AgiBot G1, evaluated over 40 trials per task.

RESEARCH EXPERIENCE (CONTINUED)

Research Assistant, Robotics Institute (continued)
Advisor: Prof. Changliu Liu

Sep 2024 – Present
Carnegie Mellon University

- **BrickCraft [8]**. Developed a visuomotor skill-composition pipeline for long-horizon brick assembly on Kinova Gen3, combining scripted picking, situated-manual visual alignment, tracking, and diffusion-policy insertion. Achieved **86.25% single-step assembly success across 240 real-world trials on 48 structural configurations**, including unseen configurations, using approximately 230 demonstrations per skill.
- **NeSyPack [2]**. Developed a neuro-symbolic bimanual task-execution pipeline combining skill graphs, object grounding, 6D pose estimation, and failure recovery; deployed on Unitree G1 and Galaxea R1 Lite. **Team award: First Prize, ICRA 2025 What Bimanual Can Do Challenge**.
- **MD-COAS [1]**. Developed a constraint-aware diffusion planner combining soft constraint penalties, hard feasibility projection, and adaptive scheduling. Achieved **100% safe-and-success rate on D3IL 7-DoF arm avoidance** across three obstacle settings (100 trajectories per setting); improved convergence and final cost on 2D planning benchmarks.
- **SPARK [3]**. Implemented and benchmarked energy-function-based humanoid safety controllers within a unified Policy–Safety architecture; validated in MuJoCo and on Unitree G1 and Galaxea R1 Lite.
- **Certifiable Semantic Robustness [9]**. Developed neuron-level metrics and Wasserstein-regularized structured pruning; achieved **96.75% verification accuracy versus 92.85% for random pruning** on CNN7 under ± 0.05 contrast perturbations at 20% channel pruning.

Research Assistant
Advisor: Prof. Ran Zhang

Jan 2023 – Sep 2024
UNC Charlotte

- **Distributed Multi-Agent Learning for Dynamic UAV Networks [4] [5]**. Developed DUCM-1/2 with heterogeneous episodic training for arbitrary UAV join and departure events; reached the **maximum achievable connectivity of 90% of users** under all three evaluated random initializations of a five-UAV network.
- **Generalization to Unseen User Distributions [6]**. Developed MA-CDQL with ResNet-based distribution encoding and continuous heatmaps; improved average connectivity by **13.4% and 18.4% over K-means** in two clustering settings, serving up to 27% more users in representative cases.

SPONSORED & COLLABORATIVE PROJECTS

ARISTOS — Interactive Assembly Training for AI-Based Reskilling

Mar 2026 – Present

Built a human-in-the-loop assembly agent converting CAD models and manuals into interactive 3D training instructions, with reusable animations for rigid-part insertion, wire routing, articulated connectors, and replay.

PerSEVE — ARM / Lockheed Martin / GE Aerospace

Sep 2025 – Jul 2026

Led an image-based Real2Sim pipeline for object reconstruction, editable scene graphs, and relation-constrained Isaac Sim synthesis; developed physics-based synthetic-data generation with domain randomization and dense annotations.

CMU Entertainment Technology Center Collaboration

May 2025 – Aug 2025

Trained and deployed RL control policies on Unitree G1; built ROS/Python interfaces connecting the robot stack to a Unity front end for interactive control and hardware prototyping.

MASCEI — ARM-Sponsored Manufacturing Automation

Oct 2024 – Sep 2025

Integrated hand–eye, tool, and environment calibration across **four industrial robot cells**, supporting a unified manufacturing calibration workflow.

TECHNICAL SKILLS

Languages	Python, MATLAB
Libraries	PyTorch, Hugging Face Transformers, Diffusers, Open3D
Robot Learning	Imitation learning, diffusion policy, (multi-agent) reinforcement learning, motion planning
Robotics	ROS, Pinocchio, LeRobot, OWLv2, FoundationPose
Simulation	Isaac Sim, Isaac Gym, MuJoCo, Gazebo
Robot Platforms	Unitree G1, Galaxea R1 Lite, AgiBot G1, Kinova Gen3